

RC-96: The S-Involution as Isospin Breaking

Research Cycle Report

Framework: 24D-DMF v8.3.5

Date: 2026-07-05

Status: CANDIDATE — PARTIALLY SUPPORTED

Preceded by: RC-95 (Turyn Z — REJECTED), RC-98 (PGL(2,23) Z — REJECTED, but discovered S)

Builds on: RC-81/82 (Non-abelian engine), RC-83 (22D perturbation), RC-90 (Unified Hamiltonian)

Executive Summary

RC-96 tested whether the involution **S**: $x \rightarrow -1/x \pmod{23}$, a confirmed element of the Golay automorphism group **M** discovered during RC-98, provides an isospin-breaking mechanism. **S** is a genuine Golay automorphism with 12 transpositions that swaps the engine's A and B orbits — exactly the structure needed for isospin-like symmetry breaking.

Critical Finding: S-perturbation achieves the **best mass-ratio match in framework history** (2.81×10^{-4} error vs 727/726), $\sim 50\times$ better than the previous best (V-only perturbation). However, the isospin interpretation remains interpretive, and the perturbation scale is still fitted, not derived.

1. The S-Involution: Definition and Properties

1.1 Definition

S: $x \rightarrow -1/x \pmod{23}$, with infinity handling:

- $S(0) = \infty \pmod{23}$ (parity bit)
- $S(23) = 0$ (cyclic origin)
- $S(x) = -x^{-1} \pmod{23}$ for $x \in \{1, \dots, 22\}$

1.2 Cycle Structure

12 transpositions, 0 fixed points:

$(0, 23), (1, 22), (2, 11), (3, 15), (4, 17), (5, 9),$
 $(6, 19), (7, 13), (8, 20), (10, 16), (12, 21), (14, 18)$

Key feature: S swaps position 0 (cyclic origin) with position 23 (parity bit) — the two “special” positions of the engine structure.

1.3 Verification: S = M

Test	Result
Preserves all 4096 codewords	✅ CONFIRMED
$S^2 = I$ (involution)	✅ CONFIRMED
Order 2	✅ CONFIRMED

2. Engine Orbit Structure and S-Action

2.1 Engine Orbits (from RC-81/82)

The multiplicative automorphism $P \ (i \rightarrow 2i \bmod 23, \text{ order } 11)$ splits the 22 non-zero positions into two cosets:

- Orbit A:** {1, 2, 3, 4, 6, 8, 9, 12, 13, 16, 18} (size 11)
- Orbit B:** {5, 7, 10, 11, 14, 15, 17, 19, 20, 21, 22} (size 11)

Position 0: fixed by P
Position 23: parity bit, fixed by all

2.2 S Swaps A B

Set	Elements	S(Image)	Result
A	{1,2,3,4,6,8,9,12,13,16,18}	{22,11,15,17,9,5,13,7,20,3,19}	= B
B	{5,7,10,11,14,15,17,19,20,21,22}	{9,13,16,2,18,21,4,6,8,1,10}	= A

S maps A → B and B → A: CONFIRMED

2.3 S-Action on Tunnel Positions

Tunnel	Position	Orbit	S(Position)	S(Orbit)
8D	8	A	20	B
7D	9	A	5	B
6D	10	B	16	A
9D	7	B	13	A

S does NOT preserve the tunnel set — maps tunnel positions to non-tunnel positions. However, the action is orbit-respecting (A→B, B→A).

3. Comparison: S vs V (Perturbation Operators)

3.1 Structural Comparison

Property	S (Involution)	V (Rank-2 Perturbation)
In M ?	Yes (code automorphism)	No (constructed operator)
Swaps AB?	Yes	Yes
**		S - V
Eigenvalue structure	11 at -1, 10 at +1, 1 near 0	Rank-2 (2 non-zero)
Physical meaning	Position permutation	Abstract orbit coupling

3.2 A/B Subspace Matrix Representation

S in A/B subspace:

```
[[ -0.957,  0.960],  
 [ 0.960, -0.957]]
```

V in A/B subspace:

```
[[ -1.833,  1.840],  
 [ 1.840, -1.833]]
```

Different structures, same qualitative behavior (swap + diagonal correction).

4. Mass Ratio Results

4.1 Direct -Scan (Best Match)

Perturbation	Best	Best Ratio	Error vs 727/726	Relative Error
V only	0.008677	1.00139203	1.46×10	0.0015%
S only	0.007513	1.00138055	2.81×10	0.000028%
V + S	0.018465	1.00137844	1.03×10	0.00010%

S-only achieves ~50× better precision than V-only.

4.2 Mechanical Response Formula (RC-86)

When the RC-86 formula $= \text{ } / (726 \times \text{split})$ is applied uniformly to all 11 degenerate pairs:

Perturbation	Formula Result	Error
V	$1453/1451 = 1.00137836$	9.48×10
S	$1453/1451 = 1.00137836$	9.48×10

Both give the **same** RC-86 ratio (because the formula normalizes away the splitting-scale difference). The direct scan shows S’s superior fine-tuning capability.

4.3 Splitting Scale Analysis

For each H degenerate pair, the S-induced first-order splitting varies:

Pair	H	S Splitting Scale	for 1/726
0	-6.458	0.720	-0.0124
1	-5.897	0.651	-0.0125
2	-4.687	0.206	-0.0313
3	-3.626	0.120	-0.0417
4	-1.196	0.035	-0.0466

Non-uniform splitting (vs V’s more uniform structure) enables better fine-tuning.

5. Group Structure

5.1 S, T where T = P (cyclic shift)

- Estimated order: ~5856 (close to $|\text{PSL}(2,23)| = 6072$)
- Large subgroup of M
- $[S, P] \neq 0$ (Frobenius norm 6.928)
- $[S, P] \neq 0$ (Frobenius norm 6.633)

5.2 S and the D Clock (RC-77)

Test	Result	
S normalizes D ?	✗ No ($S \cdot M_U \cdot S = M_U^{-1}$)	
$S^2 = I$ in symplectic rep?	✓ Yes $S, M_U >$	S, M_U
$\text{tr}(F)$	9.798	

S is **independent** of the D clock structure.

6. Honest Assessment

6.1 Confirmed Findings

Finding	Status	Evidence	
$S = M$	CONFIRMED	4096-codeword test	
S is involution (12 transpositions)	CONFIRMED	$S^2 = I$	
S swaps A B	CONFIRMED	Direct computation	
$S = V$	CONFIRMED		
S improves mass ratio	CONFIRMED	2.81×10^{-4} error $S, P >$	S, P
$0, [S, P] \neq 0$	CONFIRMED	Non-zero commutators	

6.2 Partially Supported

Finding	Status	Evidence	
S provides isospin breaking	PARTIAL	Swaps AB, but no tunnel preservation	
S has natural isospin interpretation	PARTIAL	Orbit-swap suggestive, not derived	
S mass ratio is derived	PARTIAL	still fitted, though match excellent	

6.3 Rejected

Hypothesis	Status	Reason
$S = V$ (identical)	REJECTED	Different spectral structure
S preserves tunnel positions	REJECTED	Maps tunnels to non-tunnels
S normalizes D_{clock}	REJECTED	$S \cdot M_U \cdot S = M_U^{-1}$

6.4 Open Questions

Question	Status	Next Step
Derive $\mathcal{G}$ from S -structure?	OPEN	Analyze near-zero eigenvalue (0.043)
S on proton/neutron states?	OPEN	Test state identification
S relation to D_{clock} ?	OPEN	Deeper group analysis
Cross-sector (Λ - Σ , pion)?	OPEN	Test with S -perturbation
Full group S , P , P^* ?	OPEN	BFS to verify if $\text{PSL}(2,23)$

7. Verdict

RC-96 STATUS: CANDIDATE — PARTIALLY SUPPORTED

The S -involution $S: x \rightarrow -1/x$ is a **genuine new structural element** of M that:

- 1.  **Is a code automorphism** (unlike V , which is not in M)
- 2.  **Swaps engine orbits $A \leftrightarrow B$** (the isospin-like structure)
- 3.  **Provides a NEW perturbation direction** distinct from V
- 4.  **Achieves the best mass-ratio match in framework history** (2.81×10^{-3})

However:

-  Isospin interpretation remains **interpretive**, not derived
-  Perturbation scale ϵ is **still fitted**, not derived
-  S **does not preserve tunnel positions**
-  **No cross-sector generalization** tested

Significance

S bridges the static (Golay code) and dynamic (engine) structures in a way V cannot — S is **both** a code automorphism **and** an orbit-swap operator. This dual nature makes it the most promising isospin candidate to date, even though the full derivation remains incomplete.

8. Reproducibility

All constructions use confirmed Tier A mathematics:

- Cyclic Golay code G_{24} (v8.3.5 Section 2.1)
- Engine $G = C \oplus C$ (RC-81/82)

- Orbit structure A/B (RC-81)
- 22D engine space (RC-83)

The S-involution is defined as:

```
def mobius_S(x):  
    if x == 23:          # infinity  
        return 0  
    elif x == 0:  
        return 23  
    else:  
        return (-pow(x, -1, 23)) % 23
```

Full reproduction script: RC-96_reproduction.py

9. Connection to Framework History

Cycle	Finding	Status
RC-81/82	Engine $G = C \quad C$	CONFIRMED
RC-83	Rank-2 perturbation $V =$	A B
RC-90	$= 3/\sqrt{11}$ derived	DERIVED
RC-95	Turyn Z for isospin	REJECTED (not in M)
RC-98	$PGL(2,23)$ Z	REJECTED, but discovered S
RC-96	S-involution for isospin	CANDIDATE

The S-involution was discovered as a “byproduct” of the failed RC-98 $PGL(2,23)$ hypothesis — an example of how honest rejection of one hypothesis can reveal genuinely new structure.

Report generated from direct execution of confirmed Tier A constructions.
All computations reproducible from the attached script.
No fitted parameters in structural claims.
parameter fitted in mass-ratio comparison only.

Framework Self-Description Update:

“Two nucleon-sector mechanisms. One basis-change permutation, confirmed. One candidate spin mechanism (RC-77). One newly confirmed non-abelian engine, tied to Gram matrix through Gaussian periods. One coupling coefficient derived ($= 3/\sqrt{11}$). **One new involution S M discovered, providing best mass-ratio match to date.** One scale () still fitted, with four closed avenues behind it and one new avenue (S-structure) now open.”